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Digital literacy and chemical fertilizer reduction in agriculture: a motivation-opportunity-ability perspective

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Reducing chemical fertilizer use is critical for mitigating global environmental harm and advancing sustainable agriculture across the globe. Amid digital transformation, farmers' digital literacy is a potential driver of fertilizer reduction practices, yet its underlying mechanisms remain underexplored. Based on the Motivation–Opportunity–Ability (MOA) theory, this study proposes a theoretical framework to explain the linkage between digital literacy and farmers' chemical fertilizer use decisions. Using data from 1,581 Chinese farmers collected through the China Rural Revitalization Survey (CRRS) in 2020, we construct the Generalized Ordered Logit (Gologit2) model and a standardized two-step estimation framework for mechanism analysis. Results show that: (1) Digital literacy is negatively associated with farmers' chemical fertilizer use, a finding that remains robust after conducting robustness tests and addressing endogeneity concerns. (2) This negative relationship between digital literacy and chemical fertilizer use operates through four potential pathways: higher e-commerce participation, stronger food safety awareness, greater adoption of agricultural mechanization services, and more active cooperatives participation. (3) Marginal effect analysis indicates that a one-unit increase in digital literacy is associated with a 2.3% higher probability of reducing chemical fertilizer usage and a 4.3% lower probability of increasing usage. These findings offer theoretical and practical insights for fostering green agricultural transitions through digital capacity building and provide evidence-based recommendations for sustainable agricultural policy design.

KEYWORDS
chemical fertilizer reduction, digital literacy, influence mechanism, motivation-opportunity-ability (MOA) theory, sustainable agricultural development

1 Introduction

The reduction of chemical fertilizer use is a pressing global concern. Long-term excessive use of chemical fertilizer in agriculture has led to issues such as non-point source pollution in agricultural settings and the degradation of cultivated land and agricultural products (Li et al., 2023; Chen et al., 2023). In April 2023, the UNEP (United Nations Environment Programme), WHO (World Health Organization), and FAO (Food and Agriculture Organization of the United Nations) jointly released the *Synthesis Report on the Environmental and Health Impacts of Pesticides and Fertilizers and Ways to Minimize Them: Envisioning a Chemical-safe World*. This report underscored the urgent need for transformative actions to mitigate the adverse effects of chemical fertilizer use, given their continually rising global demand. In alignment

with the United Nations' Sustainable Development Goals (SDGs), particularly Goal 2 (Zero Hunger) and Goal 15 (Life on Land), reducing chemical fertilizer use is essential for ensuring food security, promoting sustainable agriculture, and protecting ecosystems.

China is not only a major agricultural nation but also the world's largest consumer of chemical fertilizers (Ji et al., 2023). Since 2015, the country has introduced initiatives such as the "Zero Growth Action Plan for Fertilizer Use by 2020" and the "Action Plan for Fertilizer Reduction by 2025" to encourage farmers to reduce their reliance on chemical fertilizers. Data from the National Bureau of Statistics reveals that in 2022, China's chemical fertilizer usage intensity reached 298.79 kg per hectare, which is 1.33 times the international upper limit of 225 kg per hectare for safe fertilizer usage (Chen et al., 2023). This indicates that despite policy efforts, the overall usage level remains comparatively high, posing a critical issue for both scholars and policymakers.

Why is it challenging for Chinese farmers to reduce chemical fertilizer usage in agricultural production? The primary obstacles stem from a lack of motivation, opportunity, and ability. China is characterized as a "large agricultural country with smallholders", where farmers typically exhibit advanced age, limited education, and adhere to traditional farming practices (Zheng et al., 2022). Firstly, the majority of farmers are driven by the goal of maximizing profits. They often believe that higher input levels, including increased chemical fertilizer use, directly correlate with higher crop yields. Consequently, this mindset discourages them from reducing chemical fertilizer use (Zheng et al., 2020; Wang and Zhao, 2025). Secondly, from a social motivation perspective, farmers may not fully recognize the critical role that reducing chemical fertilizer use plays in ensuring food safety, environmental protection, and promoting sustainable agricultural development. Thirdly, due to limited access to information and restricted social networks, farmers frequently face difficulties in obtaining the necessary agricultural equipment, technology, and services required to reduce chemical fertilizer use effectively. Lastly, reducing chemical fertilizer use necessitates specialized knowledge and skills. Farmers often require technical support and training from professional agricultural organizations to implement sustainable practices effectively. Thus, such deficiencies in motivation, opportunity and ability collectively impede Chinese farmers from cutting their reliance on chemical fertilizers.

In the digital economy era, digital literacy, which refers to the ability to access, understand, and use digital information and technologies, plays a pivotal role in enabling farmers to access essential information and acquire necessary knowledge and skills for modern agricultural production (Choruma et al., 2024; Cheng et al., 2024). Digital literacy could foster farmers' motivation, opportunities, and abilities in agricultural production, ultimately shaping their decisions regarding chemical fertilizer use (Chandio et al., 2024). Initiatives such as China's Digital Village Construction Project have substantially improved rural internet accessibility, achieving a penetration rate of 63.8% by 2024 and engaging over 300 million rural internet users. According to the Cyberspace Administration of China, as of 2024, 50.57% of rural adults possess basic digital skills or higher, while 9.53% have achieved advanced digital proficiency. Given this context, several questions arise: Can enhanced digital literacy contribute to reducing chemical fertilizer use in rural China? If so, through what mechanisms does digital literacy influence farmers' fertilizer application decisions? These questions warrant in-depth investigation.

To address these questions, this study establishes a theoretical model based on the Motivation–Opportunity–Ability (MOA) theory to explain the influence mechanism of digital literacy on farmers' chemical fertilizer use. Using data from the China Rural Revitalization Survey (CRRS), employing a Generalized Ordered Logit (Gologit2) model and a standardized two-step estimation method to empirically test the proposed framework.

The remainder of this paper is structured as follows: Section 2 provides a comprehensive literature review and highlights the marginal contributions. Section 3 develops hypotheses and the theoretical model. Section 4 describes the data sources, empirical models, and variable selection. Section 5 presents the empirical results and discussion, and Section 6 concludes with a discussion of findings and policy implications.

2 Literature review

2.1 The concept and dimensions of digital literacy

The concept of digital literacy was first defined by Gilster (1997) as the ability to access, understand, organize, and evaluate digital information. Subsequently, authoritative organizations such as the European Commission and UNESCO (United Nations Educational, Scientific and Cultural Organization) have provided distinct definitions (Cheng et al., 2024). The European Commission describes digital literacy as the confident, critical, and creative use of digital technologies to achieve goals related to work, learning, leisure, and societal participation (Ferrari, 2013; Puigarnau, 2018; Mattar et al., 2022). UNESCO defines it as the ability to safely and appropriately acquire, manage, understand, integrate, communicate, evaluate, and create information through digital technology. In China's "Program for Action to Improve National Digital Literacy and Skills", digital literacy encompasses a range of qualities and abilities, including digital acquisition, production, use, evaluation, interaction, sharing, and innovation.

In this study, digital literacy is conceptualized as a transversal key competence for farmers, enabling them to access, acquire, and utilize digital information, thereby facilitating the development of other competencies such as language proficiency, lifelong learning skills, and cultural awareness. We structure digital literacy around three core dimensions: digital access literacy, digital acquisition literacy, and digital application literacy. Digital access literacy refers to farmers' possession of digital devices (e.g., computers and smartphones) and stable internet access. Digital acquisition literacy pertains to farmers' proficiency in using these devices to effectively gather needed information. Digital application literacy involves the ability to communicate online and apply digital information to address challenges in agricultural production.

2.2 The relationship between digital literacy and farmers' production decision-making

A growing body of literature has examined the role of digital literacy in shaping farmers' production decision-making, particularly those related to environmental sustainability. Existing studies suggest that digital literacy significantly enhances the adoption of green

agricultural practices (Li et al., 2024). For instance, Wang and Long (2025) demonstrated that farmers' digital literacy significantly promotes the green transition of food production by increasing service outsourcing participation and improving green cognition. Similarly, Liu et al. (2025) demonstrated that digital literacy promotes the adoption of green production technologies by strengthening risk perception, digital social capital, and the effectiveness of technology extension services. Further supporting this, Liu et al. (2024) illustrated that digital literacy positively influences farmers' green production behavior by enhancing their ecological cognition. These findings are consistent with broader evidence that digital literacy fosters pro-environmental behaviors among farmers through improved policy awareness (Chen et al., 2021) and strengthened subjective norms, behavioral attitudes, and perceived behavioral control (Lu et al., 2024). Grant et al. (2026) show that UK farmers exhibit significant "algorithm aversion" in fertilizer decision-making, with barriers including cost, poor usability, low trust, and unfamiliar metrics. Kumar et al. (2026) identify high upfront capital costs, data privacy concerns, and a lack of farmer digital literacy as significant obstacles to adopting digital agricultural technologies. Sarku and Kranjac-Berisavljevic (2025) find that farmers' access to digital technologies in Ghana is shaped by intersecting factors such as gender, age, education, income, religion, and geography, leading to unequal adoption. Beyond production practices, digital literacy also contributes to job quality (Zhang, 2025), entrepreneurial behavior (Cheng et al., 2024), and green consumption patterns (Hu and Meng, 2023), underscoring its multifaceted impact in rural contexts.

Concurrently, studies focusing on digital technology adoption have widely documented its significant role in optimizing fertilizer use. Theoretically, digital technology adoption has a clear distinction from digital literacy: digital technology adoption refers to farmers' observable behavioral use of digital tools, such as internet use, e-commerce, and ICTs, in agricultural production, while digital literacy is farmers' intrinsic, multidimensional capability to access, acquire, and apply digital information for making scientific production decisions. Research within China demonstrates that digital tools enhance input efficiency and promote sustainability (Qiu et al., 2025). For example, Zhu et al. (2023) showed that information and communication technologies (ICTs) improve fertilizer use efficiency in grain production. Ma and Zheng (2022) highlighted the heterogeneous impacts of information technology on reducing fertilizer expenditures among wheat farmers. Internet use, in particular, facilitates organic fertilizer adoption by expanding social networks for knowledge sharing (Ma et al., 2022). E-commerce platforms further encourage green technology adoption, including organic fertilizers (Qiu et al., 2024), with macro-level evidence indicating a 21% reduction in chemical fertilizer use under e-commerce policy interventions (Ji et al., 2023).

2.3 The MOA theory and its application

The MOA theory posits that a behavior is more likely to be performed when an individual (1) perceives the behavior as beneficial and is aware of the consequences of inaction (Motivation); (2) has viable and accessible options that facilitate the behavior (Opportunity); and (3) possesses the necessary skills and knowledge to execute the behavior (Ability) (van Geffen et al., 2020). Originating from research on information processing in communication and marketing (Batra and Ray, 1986; Andrews, 1988; MacInnis and Jaworski, 1989), the MOA framework has been widely applied in public management, environmental management, and knowledge management due to its strong explanatory power and predictive validity (Ölander and Thøgersen, 1995; Siemsen et al., 2008; Baumhof et al., 2018; Cui et al., 2020; Guenzi and Nijssen, 2023).

For instance, motivation, opportunity, and ability positively affect consumers' readiness to engage in circular food consumption (Raimondo et al., 2024). Awareness interventions can improve motivation and ability, leading to reduced household food waste (Soma et al., 2021). In the American workplace, opportunity has the strongest effect on energy-saving behaviors, followed by motivation and ability (Li et al., 2019).

2.4 Research gap and marginal contribution

While existing literature has considerably advanced our understanding of the relationship between digital technology use and farmers' production decision-making, several critical research gaps remain, which this study aims to address.

First, although prior studies have established a positive correlation between digital technology adoption, such as internet use, e-commerce, and ICTs, and the reduction in fertilizer application (e.g., Zhu et al., 2023; Qiu et al., 2024; Ji et al., 2023), most focus on the role of technology access or usage per se, rather than on farmers' intrinsic digital literacy. Digital literacy, as a multidimensional capability encompassing access, acquisition, and application of digital information, represents a deeper and more foundational competence than mere technology adoption. Yet, its specific influence on chemical fertilizer use has not been systematically investigated.

Second, while several scholars have explored the impact of digital literacy on broader pro-environmental behaviors or the adoption of green production technologies (e.g., Li et al., 2024; Liu et al., 2025; Chen et al., 2021), the mechanism through which digital literacy influences farmers' decision-making regarding chemical fertilizer reduction remains underexplored. Most studies lack a coherent theoretical framework to unpack the mediating pathways, especially from a behavioral perspective that integrates motivational, opportunistic, and capacitating factors.

Third, although the MOA framework has been widely applied in environmental and consumer behavior research, its application in the context of farmers' agricultural production decisions, particularly in developing countries like China, is still novel. To the best of our knowledge, no prior study has employed the MOA theory to analyze how digital literacy influences farmers' chemical fertilizer use through improved motivation, opportunity, and ability.

Therefore, this study contributes to the literature in the following ways: (1) by shifting the focus from digital technology use to farmers' digital literacy as a core determinant of chemical fertilizer reduction; (2) by proposing and testing an integrated theoretical model based on the MOA framework to elucidate the mediating mechanisms, namely, e-commerce participation (economic motivation), food safety awareness (social motivation), agricultural mechanization services adoption (opportunity), and cooperatives participation (ability); and (3) by providing empirical evidence from a large-scale survey in China, using robust econometric methods to establish causality and validate the proposed pathways.

3 Theoretical analysis

3.1 The influence of farmers' digital literacy on chemical fertilizer use

Farmers often operate under significant information asymmetry in agricultural input markets, particularly regarding the quality, appropriate application methods, and long-term impacts of chemical fertilizers (Ullah et al., 2020). This information gap, combined with limited technical competence, frequently leads to inefficient and excessive fertilizer use. The theoretical linkage between digital literacy and input use efficiency can be grounded in information economics and resource-based theory. Digital literacy serves as a valuable resource that enables farmers to access, process, and utilize information more effectively, thereby reducing information asymmetry and improving decision-making quality.

Within the MOA framework, digital literacy primarily boosts farmers' ability to make informed decisions about chemical fertilizer use. Specifically, digital literacy—through its dimensions of access, acquisition, and application—empowers farmers to obtain technical knowledge about optimal fertilizer application rates, understand the environmental and health implications of overuse, and learn about alternative practices and technologies. This aligns with the concept of informational capability in development economics, which emphasizes how information access can enhance human agency and improve economic decisions (Choruma et al., 2024).

Moreover, the relationship is supported by technology acceptance models that posit that individuals' use of technology is influenced by perceived usefulness and ease of use—both of which are heightened by greater digital literacy. When farmers can effectively use digital tools, they are more likely to access agricultural extension services, engage with online training modules, and utilize decision-support systems that recommend optimal fertilizer application rates (Yang et al., 2024). Thus, digital literacy reduces cognitive and technical barriers to the adoption of sustainable practices. Therefore, we hypothesize Hypothesis 1 (H1).

> *H1*: The improvement of digital literacy is negatively associated with farmers' chemical fertilizer use.

3.2 Theoretical analysis of how farmers' digital literacy reduces chemical fertilizer use

Coherent with Ölander and Thøgersen (1995) as well as Soma et al. (2021), this study expects that MOA might successfully interpret the influence mechanism of digital literacy on chemical fertilizer use due to the following two main reasons: (1) the accuracy of predicting behaviors with environmental implications enhances when considerations are given to both ability and opportunity, and (2) the MOA is particularly well-suited for anticipating behaviors associated with goals that necessitate discipline, ability, and pertinent resources in this context.

3.2.1 Analysis of influence mechanism from an economic motivation perspective

By facilitating farmers' e-commerce participation, digital literacy can contribute to the economic motivation for reducing chemical fertilizer use. E-commerce platforms enable direct farmer–consumer interaction, giving farmers clearer signals of market premium for green products (Zhou and Duan, 2022; Li et al., 2021). Consequently, farmers are motivated to lower their chemical fertilizer use to receive positive feedback and higher returns from consumers. Moreover, the reduced role of intermediaries strengthens farmers' bargaining ability, further incentivizing the production of green agricultural products (Qiu et al., 2024). In a word, farmers with high digital literacy are more likely to engage with e-commerce platforms, and their participation in these platforms increases their motivation to reduce chemical fertilizer use. Hypothesis 2 (H2) is proposed.

> *H2*: The improvement of digital literacy is negatively correlated with farmers' chemical fertilizer use through enhancing their e-commerce participation.

3.2.2 Analysis of influence mechanism from a social motivation perspective

By improving food safety awareness, digital literacy tends to strengthen farmers' social motivation to pursue fertilizer reduction. In response to national strategies like the "Healthy China" and "The beautiful countryside", the government departments publicize online information about the health and environmental risks of excessive chemical fertilizer use. These include extensively publicizing information via the internet that excessive chemical fertilizer use could cause both unhealthy agricultural food and environmental pollution. Farmers with high digital literacy can access related information through videos, texts, and images, fostering an emotional connection to environmental issues and raising awareness of food risks (Chen et al., 2021; Liu et al., 2024). This can enhance their motivation to reduce chemical fertilizer use in agricultural practices (Qu et al., 2024). Hypothesis 3 (H3) is proposed.

> *H3*: The improvement of digital literacy is negatively linked to farmers' chemical fertilizer use through raising their food safety awareness.

3.2.3 Analysis of influence mechanism from an opportunity perspective

Digital literacy may enhance farmers' opportunities to reduce chemical fertilizer use by promoting access to agricultural mechanization services. Farmers with advanced digital literacy can access online platforms to overcome information barriers and access agricultural mechanization services conveniently. These services provide uniformity, standardized procedures, and precision in fertilizer application, reducing subjective overuse and lowering application intensity (Su et al., 2022). By equipping farmers with the necessary equipment and technology, agricultural mechanization services offer them the opportunities to reduce the chemical fertilizer use objectively, thereby finally decreasing their chemical fertilizer use intensity. Hypothesis 4 (H4) is proposed.

> *H4*: The improvement of digital literacy has a negative relationship with farmers' chemical fertilizer use through promoting their adoption of agricultural mechanization services.

3.2.4 Analysis of influence mechanism from an ability perspective

By facilitating farmers' participation in cooperatives, digital literacy strengthens their capacity to reduce chemical fertilizer

application. The core of the ability dimension lies in the process through which farmers transform external resources into their own knowledge, skills, and capacity for action (Ölander and Thøgersen, 1995). Cooperatives, as voluntary agricultural organizations collectively owned and managed by farmers, provide critical support in the form of agricultural supplies and technical training (Lei et al., 2024). Farmers with higher digital literacy are better able to access information about the benefits of cooperatives and are more likely to participate actively (Wang et al., 2024). Through continuous participation, farmers internalize collective resources and training support into practical production skills—a process of internalization that epitomizes capacity building (Lei et al., 2024; Mei et al., 2024). Accordingly, Hypothesis 5 (H5) is proposed.

> *H5*: The improvement of digital literacy is negatively associated with farmers' chemical fertilizer use by increasing cooperative participation.

In summary, the theoretical model of digital literacy influences chemical fertilizer use is illustrated in Figure 1.

4 Data and modeling

4.1 Data source

The data used in this study are derived from the China Rural Revitalization Survey (CRRS). This survey was organized by the Rural Development Institute Chinese Academy of Social Sciences in 2020. Using the principle of multi-stage stratified random sampling, the CRRS sample covered 10 provinces (Guangdong, Zhejiang, Shandong, Anhui, Henan, Heilongjiang, Guizhou, Sichuan, Shaanxi Province and Ningxia Hui Autonomous Region), 50 counties, 156 townships, 308 administrative villages and 3,833 farm households in China. This survey focuses on major issues in rural revitalization, such as farmers' income and expenditure, agricultural production structure, mode and input, production environment, informatization and e-commerce development, covering all the micro indicators needed in this paper. The comprehensive dataset facilitates an in-depth analysis of the correlation between digital literacy and farmers' decision-making regarding chemical fertilizer use. In addition, the survey strictly adhered to the principle of informed consent during data collection. All participating farm households were informed of the survey purpose, content, and data usage prior to participation. Besides, this dataset has been widely used by scholars to study the relationship between rural revitalization, information technology usage, and farmers' production behavior (Wang et al., 2024; Li and Gao, 2025).

This study follows a series of basic principles when constructing research data to ensure the rigor and reliability of the research results. Firstly, the head of the household is the core member of the family's decision-making on agricultural production activities in rural areas of China (Yu et al., 2023), and the decision-making on fertilizer use of farmer households is generally determined by the head of the household. Therefore, in order to ensure the independence of data analysis at the family level, this study only retains the information of each household head, thereby eliminating potential data interference from other family members (Wang et al., 2024). Secondly, in order to ensure data quality, samples lacking key variable information are also excluded. Through these rigorous screening and processing steps, we obtained a final dataset containing 1,581 farmer households.

4.2 Models

4.2.1 The benchmark regression model

In this study, farmers' chemical fertilizer use decisions (1 = reduction, 2 = unchanged, 3 = increase) represent an ordinal categorical variable. Both ordered logit (Ologit) and Gologit2 models are suitable for empirical estimation. The Ologit requires fulfillment of the proportional odds assumption. However, likelihood ratio tests indicate that key independent variables—digital literacy ($p = 0.046$) and digital access literacy ($p = 0.036$)—violate this assumption. This suggests that the influence parameters of independent variables on fertilizer use decisions are not constant across ordinal categories, rendering the Ologit inappropriate. The Gologit2 relaxes the proportional odds assumption by allowing independent variables to exert differential effects across outcome categories while preserving the ordinal nature of the dependent variable (Williams, 2006; Johnston et al., 2020). Therefore, this study employs the Gologit2, specified as follows (Equation 1):

$$P\left(Y_i > j\right) = \frac{\exp\left(\alpha_j + \beta_j X_i + \gamma_j Control_i\right)}{1 + \exp\left(\alpha_j + \beta_j X_i + \gamma_j Control_i\right)},\ j = 1,2 \qquad (1)$$

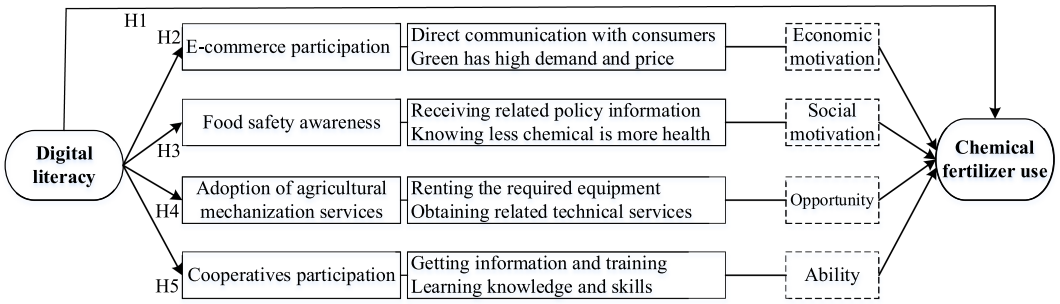


FIGURE 1
Theoretical model of how digital literacy reduces chemical fertilizer use.

Let Y_i denote the fertilizer use decision category of farmer i, X_i represent core independent variables including digital literacy, digital access literacy, digital acquisition literacy, and digital application literacy, and $Control_i$ indicate the control variables. The parameters α_j are threshold points, while β_j and γ_j represent category-specific parameter vectors. For each threshold j the model estimates the cumulative probability $P(Y_i > j)$, where $j=1$ corresponds to the transition from "reduction" to "unchanged or increase", and $j=2$ indicates the transition from "reduction or unchanged" to "increase." Here, β_1 captures how independent variables influence farmers' transition away from the "reduction" decision, whereas β_2 reflects their impact on adopting the "increase" decision. Probabilities for each outcome category can be derived as follows (Equation 2):

$$P(Y_i=1)=1-P(Y_i>1)$$

$$P(Y_i=2)=P(Y_i>1)-P(Y_i>2)$$

$$P(Y_i=3)=P(Y_i>2) \quad (2)$$

4.2.2 Mechanism test models

Guided by the theoretical framework, this study adopted the standard two-step mediation framework widely used in observational and causal inference studies (Jiang, 2022; Li and Gao, 2024). This method identifies the causal chain through statistically significant linkages and avoids potential endogeneity bias inherent in traditional multiple-step mediation approaches. The core logic of the two-step approach is as follows.

In the first step, identify the causal effect of the core independent variable on each mediating variable. Apply appropriate regression models according to the measurement scale of each mediator.

For binary mediating variables (e-commerce participation M_{1i}, cooperatives participation M_{4i}), we use binary probit models (Equation 3):

$$P(M_{ki}=1)=\Phi(\gamma_{0k}+\gamma_{1k}X_i+\gamma_{2k}Control_i), k=1,4 \quad (3)$$

For the ordered mediating variable (food safety awareness M_{2i}), we use an ordered logit model (Equation 4):

$$P(M_{2i}>m)=\frac{\exp(\nu_{0m}+\nu_1X_i+\nu_2Control_i)}{1+\exp(\nu_{0m}+\nu_1X_i+\nu_2Control_i)}, m=1,2,3,4 \quad (4)$$

For the continuous mediating variable (adoption of agricultural mechanization services M_{3i}), we use an OLS regression model (Equation 5):

$$M_{3i}=\pi_0+\pi_1X_i+\pi_2Control_i+\varepsilon_{3i} \quad (5)$$

In the second step, verify the total causal effect of digital literacy on farmers' fertilizer use decisions, which is consistent with the benchmark regression results based on the Gologit2 framework.

Where M_{1i} denotes e-commerce participation, M_{2i} denotes food safety awareness, M_{3i} denotes agricultural mechanization services adoption, M_{4i} denotes cooperatives participation; $\gamma_{0k}, \nu_{0m}, \pi_0$ denote constant terms; $\gamma_{1k}, \nu_1, \pi_1$ denote the coefficients of independent variables; $\gamma_{2k}, \nu_2, \pi_2$ denote coefficients of control variables; and ε_{3i} denotes the random error term.

4.3 Variable selection

4.3.1 Core independent variable

The core independent variables are digital literacy, and its 3 subdimensions: digital access literacy, digital acquisition literacy, and digital application literacy. Factor analysis is employed to measure these three sub-dimensions. Each sub-dimension is measured by two specific indicators, all of which are derived from self-reported questions in the CRRS questionnaire. These three sub-dimensions collectively constitute the overall construct of digital literacy. The complete index system is presented in Table 1.

Factor analysis can extract latent factors from multiple variables, facilitating dimensionality reduction of multivariate data. Since each dimension in this paper comprises multiple measurement items, factor analysis is an appropriate method for dimension reduction. Before performing factor analysis, the Bartlett's Test of Sphericity and Kaiser-Meyer-Olkin (KMO) test were conducted to verify its adequacy. The *p*-values of Bartlett's Test of Sphericity for digital literacy, digital access literacy, digital acquisition literacy, and digital application literacy are <0.001, 0.054, <0.001, <0.001, respectively, and the KMO values are 0.533, 0.500, 0.500, 0.500, respectively. These results collectively satisfy the conditions for conducting factor analysis. The factor analysis extracted one common factor for each dimension respectively, with cumulative variance contribution rates ranging from 51.33% to 75.49%, indicating good structural validity of the indicators.

4.3.2 Dependent variable

The dependent variable is farmers' decision-making on chemical fertilizer use, whose value is measured against the responses to the CRRS questionnaire question "What is your status of chemical fertilizer use compared to 5 years ago: 1 = Decrease; 2 = Unchanged; 3 = Increase?". Table 1 shows that the mean value of chemical fertilizer use is 2.497, suggesting that there is a significant increase trend in chemical fertilizer use among the respondents. Table 2 shows the definitions of the variables.

4.3.3 Control variables

Drawing on existing studies (Ma and Zheng, 2022; Qiu et al., 2024), this study selects control variables from three dimensions: individual characteristics, household characteristics, and village characteristics. Individual characteristics consist of six variables: gender, age, educational level, employment status, marital status, and role in village. Household characteristics include five variables: land scale,

degree of land fragmentation, household labor inputs, annual household income (Taking logarithmic), and agricultural insurance availability. For village characteristics, variables such as village landforms and village location are considered.

4.3.4 Mechanism variables

Based on the theoretical model proposed in Section 3, the mechanism variables are e-commerce participation, food safety awareness, the adoption of agricultural mechanization services, and cooperatives participation. They are measured against responses to the CRRS questionnaire questions "Whether your family operates products traded through the internet", "Do you care about food safety", "The proportions of mechanical operations" minus "The proportions of in-house mechanical operations", and "Whether your family is a member of cooperatives" respectively.

5 Results and discussion

5.1 Results of benchmark regression

Table 3 presents the benchmark regression results of the influence of digital literacy on chemical fertilizer use. The mean VIF across models is well below the 10 threshold and close to 1, ruling out substantial multicollinearity. The Pseudo R^2 values are consistent with the common range of Gologit2 in the field of agricultural economics, indicating that the models possess reasonable explanatory power.

The results from column (1) indicate a significant negative association between digital literacy and chemical fertilizer use at the 10% statistical level, suggesting that higher levels of digital literacy dampen farmers' willingness to use chemical fertilizers and thus drive their decisions to reduce fertilizer application. H1 is thus verified. This finding aligns with the informational capability theory in development economics (Choruma et al., 2024), which posits that enhanced digital competence allows farmers to access, evaluate, and apply agronomic information more effectively, thereby mitigating information asymmetry and optimizing input decisions. Furthermore, coefficient values indicate that a one-unit increase in digital literacy corresponds to a 0.358 reduction in the log-odds of transitioning from "reduction" to "unchanged or increased" fertilizer use, and it reflects a 0.183 reduction in the log-odds of shifting from "reduction or unchanged" to "increased" application. This pattern suggests that digital literacy has a more pronounced dampeing impact on farmers' decisions to increase chemical fertilizer use relative to other decision-making pathways.

As shown in column (3), digital acquisition literacy exhibits a robust negative correlation with chemical fertilizer use, showing statistical significance at the 1% level in the Y > 1 equation and at the 5% level in the $Y > 2$ equation. This pattern underscores the critical role of farmers' effective digital information acquisition. It supports the argument that the ability to proactively obtain digital resources, such as technical guidelines on fertilization or market demand for green products, directly challenges misconceptions like "more fertilizers guarantee higher yield" (Zheng et al., 2022), thereby driving farmers to make informed and rational agricultural input decisions.

The columns (2) and (4) indicate that neither digital access literacy nor digital application literacy shows a statistically significant impact on farmers' fertilizer use decision-making. This resonates with Ma and Zheng (2022), who observed heterogeneous impacts of IT adoption, where usage efficacy rather than superficial engagement drives agricultural behavioral change. Plausible explanations may account for these observed non-significant relationships. Regarding digital access literacy, its non-significance may stem from the widespread penetration of basic digital infrastructure in the study region. When internet connectivity and smart device ownership become nearly universal, variation in access literacy among farmers decreases significantly. Consequently, this factor no longer acts differentiates agricultural management decisions. Besides, the non-significant coefficient for digital application literacy could reflect a disconnect between farmers' technical proficiency and their agricultural practices. Farmers may possess adequate skills to operate digital applications for general purposes such as social communication or entertainment, yet fail to transfer these competencies to agricultural production contexts. This compartmentalization of digital skills suggests that mere technical ability does not automatically translate into causal capacity for specialized agricultural management.

TABLE 1 Measurement of digital literacy and three sub-dimensions.

Dimension	Indicators	Mean	SD
Digital literacy	Factor analysis composite score	0	0.740
Digital access literacy	Factor score for digital access	0	1
	Do you have internet-enabled devices at home: 0 = No, 1 = Yes	0.997	0.050
	Do you have a high-quality internet connection at home: 0 = No, 1 = Yes	0.481	0.500
Digital acquisition literacy	Factor score for digital acquisition literacy	0	1
	Are you able to access relevant information anytime through your phone or the internet on your own if you have a daily need: 0 = No, 1 = Yes	0.552	0.497
	Are you able to use the features of a 4G/5G phone without any difficulties: 0 = No, 1 = Yes	0.534	0.499
Digital application literacy	Factor score for digital application literacy	0	1
	Have you ever engaged in communication regarding significant public affairs with the village via a WeChat group: 0 = No, 1 = Yes	0.700	0.459
	Whether the information you get through the Internet can fully satisfy your daily needs: 0 = No, 1 = Yes	0.371	0.483

TABLE 2 The definitions and descriptive statistics of variables.

Variable type	Variable name	Definitions	Mean	SD
Dependent variables	Chemical fertilizer use	What is your status of chemical use compared to 5 years ago: 1 = Decrease; 2 = Unchanged; 3 = Increase	2.497	0.627
Core independent variable	Digital literacy	Factor analysis composite score	0	0.740
	Digital access literacy	Factor score for digital access	0	1
	Digital acquisition literacy	Factor score for digital acquisition	0	1
	Digital application literacy	Factor score for digital application	0	1
Individual characteristics	Gender	0 = Male, 1 = Female	0.046	0.209
	Age	Age	53.082	9.697
	Educational level	1 = Not attended school; 2 = Primary school; 3 = Junior high school; 4 = high school; 5 = Vocational school; 6 = Secondary vocational-technical school; 7 = college; 8 = Undergraduate; 9 = Postgraduate	2.820	0.993
	Employment status	1 = Full-time farming; 2 = Non-farm employment; 3 = Part-time (farm and non-farm)	1.810	0.997
	Marital status	1 = Married; 2 = Unmarried; 3 = Divorced; 4 = Widowed	1.116	0.537
	Role in village	1 = Ordinary villagers; 2 = Village secretary; 3 = committee member; 4 = Team leader; 5 = Cooperatives or collective economic organization board member	1.581	1.261
Household characteristics	Land scale	Total cultivated land area managed by farmers (Mu)	2.588	1.266
	Degree of land fragmentation	Number of land/ Land scale	0.923	1.347
	Household labor inputs	Labor invested in production activities (Person)	2.904	1.110
	Annual household income	Annual household income	10.657	1.515
	Agricultural insurance availability	0 = No, 1 = Yes	0.467	0.491
Village characteristics	Village landforms	1 = plain; 2 = hills; 3 = mountainous and semi-mountainous areas	1.881	0.886
	Village location	Is it a suburban area of the city: 0 = No, 1 = Yes	0.180	0.383
Mechanism variables	E-commerce participation	0 = No, 1 = Yes	0.047	0.211
	Food safety awareness	1 = very unconcerned; 2 = comparatively unconcerned; 3 = generally; 4 = comparatively concerned; 5 = very concerned	4.322	0.886
	Adoption of agricultural mechanization services	Volume of agricultural mechanization services purchased by farmers	0.155	0.357
	Cooperatives participation	0 = No, 1 = Yes	0.255	0.436

5.2 Results of robustness test and endogeneity test

5.2.1 Results of robustness test

To ensure the reliability of the benchmark regression results, robustness tests are conducted using alternative model specifications and sample treatments. Specifically, the models are re-estimated using the Ordered Logit (Ologit) and Ordered Probit (Oprobit) models, and the bilateral truncation method is applied to address potential outliers. As presented in Table 4, the coefficient of digital literacy remains significantly negative across all alternative specifications, confirming the robustness of the benchmark regression results.

5.2.2 Results of endogeneity test

The benchmark regression may suffer from endogeneity arising from reverse causality and omitted variables. For instance, farmers who are more inclined to adopt sustainable practices might also proactively improve their digital skills, leading to reverse causality. To establish a more robust causal inference for the association between digital literacy and chemical fertilizer use, this study employs an instrumental variable (IV) approach, specifically a two-stage estimation framework based on the control function method—a standard two-stage approach for nonlinear ordered choice models. In this study, the provincial Internet penetration rate is employed as the instrumental variable (Zhang et al., 2020; Chen and Zhao, 2021; Feng et al., 2025). This variable is suitable as it directly influences farmers' digital literacy but does not have a direct effect on their chemical fertilizer use, thereby satisfying the correlation and exogeneity criteria for instrumental variables. Considering that the dependent variable is an ordered categorical variable, which matches the benchmark regression, this study uses the Instrumental Variable Generalized Ordered Logit (IV-GOLOGIT2) model for estimation.

The estimation results support the validity of this approach. As shown in Table 5, the first-stage results indicate a statistically significant and positive effect of the instrumental variable on digital literacy. The F-statistic of 16.31 comfortably exceeds the Staiger and Stock (1997) rule-of-thumb threshold of 10, ruling

TABLE 3 Benchmark regression results of influence of digital literacy on chemical fertilizer use.

Variables	(1)		(2)		(3)		(4)	
	(Y > 1)	(Y > 2)	(Y > 1)	(Y > 2)	(Y > 1)	(Y > 2)	(Y > 1)	(Y > 2)
Digital literacy	−0.358* (0.158)	−0.183* (0.077)						
Digital access literacy			−0.122 (0.135)	−0.094 (0.060)				
Digital acquisition literacy					−0.291** (0.115)	−0.130** (0.056)		
Digital application literacy							−0.046 (0.104)	0.001 (0.054)
Controlled variables	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Pseudo R^2	0.041		0.038		0.041		0.0304	
Mean VIF	1.18		1.17		1.18		1.17	
N	1,581							

*, **, and *** indicate significance at the 10, 5, and 1% statistical levels, respectively, with standard errors in parentheses. Similarly hereinafter.

TABLE 4 Results of robustness test.

Variables	Ologit model	Oprobit model	Bilateral truncation	
			(Y > 1)	(Y > 2)
Digital literacy	−0.206*** (0.075)	−0.124*** (0.044)	−0.513** (0.179)	−0.182* (0.090)
Digital access literacy	−0.096 (0.059)	−0.054* (0.033)	−0.153 (0.156)	−0.113 (0.080)
Digital acquisition literacy	−0.150** (0.055)	−0.092*** (0.033)	−0.379** (0.128)	−0.120* (0.060)
Digital application literacy	−0.010 (0.053)	−0.007 (0.031)	−0.090 (0.115)	0.028 (0.061)
Control variables	Controlled	Controlled	Controlled	
Pseudo R^2	0.029–0.032	0.030–0.033	0.042–0.046	
N	1,581	1,581	1,444	

TABLE 5 Result of endogeneity test.

Variables	Digital literacy	Chemical fertilizer use
Provincial level of internet access	0.060*** (0.023)	
Digital literacy(predicted)		−5.896*** (1.179)
Digital literacy(residual)		−0.188** (0.075)
Control variables	Controlled	Controlled
F	16.31	
Pseudo R^2		0.041
Log likelihood		−1330.933
LR ×2		112.40***
N	1,581	1,581

out concerns regarding a weak instrument. In the second stage, the instrumented digital literacy continues to exhibit a significantly negative effect on chemical fertilizer use, with a substantially larger magnitude than the baseline regression. A formal endogeneity test—such as the significance of the residual inclusion term—confirms the presence of endogeneity in the baseline specification. This result implies that the initial estimates were inconsistent and likely underestimated the true effect of digital literacy, thereby rendering the IV-corrected results more reliable.

5.3 Results of mechanism test

Table 6 presents the results of the mechanism tests, which elucidate the pathways through which digital literacy is associated with

TABLE 6 Mechanism test: the effect of digital literacy on mediating variables.

Variables	E-commerce participation	Food safety awareness	Adoption of agricultural mechanization services	Cooperatives participation
Digital literacy	0.168* (0.087)	0.320*** (0.070)	0.028* (0.015)	0.232*** (0.051)
Control variable	Controlled	Controlled	Controlled	Controlled
Constant term	−2.387*** (0.637)		0.607*** (0.121)	−1.246*** (0.412)
Pseudo R^2	0.080	0.017		0.023
R^2			0.144	
N	1,493	1,581	1069	1,575

A small number of samples were removed due to the absence of the mechanism variables, so the sample size of the dataset was less than or equal to 1581.

TABLE 7 Marginal effect results.

Variables	Decrease	Unchanged	Increase
Digital literacy	0.023* (0.010)	0.020 (0.018)	−0.043* (0.018)
Control variable	Controlled	Controlled	Controlled

farmers' chemical fertilizer use. Following the standard two-step approach for mediating effect analysis in causal inference research (Jiang, 2022), this study tests the theoretical mechanism based on the MOA framework. This two-step mediation framework is consistent with the benchmark generalized ordered logit model, ensuring estimation consistency for the ordered dependent variable. The core logic of the two-step approach is: first, identify the causal effect of the core independent variable on the mediating variables; second, verify the total causal effect of digital literacy on farmers' fertilizer use decision—consistent with the benchmark regression results. For each mediating variable, its impact on fertilizer use is supported by solid theoretical logic and existing literature.

The result in column (1) indicates that digital literacy is positively correlated with e-commerce participation (coefficient = 0.168, $p < 0.1$), which in turn may contribute to reduced chemical fertilizer use. This aligns with the economic motivation perspective, as e-commerce platforms facilitate direct farmer-consumer interactions, enabling farmers to perceive market premiums for green products and thus incentivizing them to reduce chemical inputs (Zhou and Duan, 2022; Qiu et al., 2024). Consistent with evidence from Sub-Saharan Africa, e-commerce platforms drive improvements in market access, facilitate information flow, and encourage the adoption of sustainable agricultural practices (Morepje et al., 2024). Farmers with higher digital literacy are more likely to engage in e-commerce, access real-time market information, and respond to consumer preferences for environmentally friendly products, thereby driving them to reduce fertilizer use to meet market demands (Li et al., 2021).

Column (2) shows that digital literacy is robustly positively linked to greater food safety awareness (coefficient = 0.320, $p < 0.01$), which plays a mediating role in its negative relationship with chemical fertilizer use. This finding supports the social motivation pathway within the MOA framework, as digital literacy enables farmers to access and comprehend information regarding the health and environmental risks associated with excessive fertilizer use (Chen et al., 2021; Liu et al., 2024). Through diverse digital channels—such as online videos, news articles, and public service announcements—farmers gain heightened awareness of the benefits of sustainable agricultural practices. This enhanced consciousness motivates them to adopt greener production methods and reduce reliance on chemical inputs (Qu et al., 2024; Zeng et al., 2020), thereby translating digital access into tangible causal impacts on agricultural behavioral change.

In column (3), digital literacy is linked to a significant increase in the adoption of agricultural mechanization services (coefficient = 0.028, $p < 0.1$), a change that may subsequently correspond to lower chemical fertilizer use. This reflects the opportunity mechanism, as digital platforms help mitigate information asymmetries in the supply and demand of mechanization services (Su et al., 2022). Agricultural mechanization services often incorporate precision fertilization technologies, such as soil testing and variable rate application, which optimize input use and minimize waste (Yang et al., 2024). Farmers with higher digital literacy can better identify, evaluate, and access these services, thereby reducing their reliance on subjective and often excessive fertilizer application through a causal pathway.

Column (4) demonstrates that digital literacy has a strong positive association with cooperatives participation (coefficient = 0.232, $p < 0.01$), which facilitates farmers' capacity to reduce chemical fertilizer use through an underlying mechanism. Cooperatives provide technical training, shared resources, and peer learning opportunities that are critical for adopting sustainable practices (Lei et al., 2024; Mei et al., 2024). As in Ghana, membership in farmer-based organizations increases the adoption digital agricultural solutions (Miine et al., 2023). Digital literacy facilitates access to information about cooperatives' benefits and improves communication among members, strengthening collective action and knowledge diffusion (Wang et al., 2024). This finding underscores the causal role of cooperatives in building farmers' capacity to implement green production techniques.

5.4 Results of marginal effect

In the Gologit2 model, the marginal effect represents the change in probability of an individual selecting a specific outcome category resulting from a unit change in an independent variable, while holding other variables constant. Further analysis of the marginal effects provides valuable insights into how variations in farmers' digital literacy

influence the probability distribution across different chemical fertilizer use decisions. The Average Marginal Effects (AMEs) method is employed to calculate the mean marginal effects across all observations, with the results presented in Table 7.

The marginal effect results reveal a nuanced and asymmetric influence of digital literacy on farmers' decision-making of chemical fertilizer use. Specifically, a one-unit increase in digital literacy is associated with a 4.3% decrease in the probability of increasing chemical fertilizer use, which remains larger in magnitude than the 2.3% increase in the probability of decreasing usage. This asymmetry suggests that digital literacy deters further intensification of fertilizer application more effectively rather than as an immediate driver of active reduction. This pattern aligns with the behavioral inertia commonly observed in smallholder agriculture, where farmers are often reluctant to deviate from established practices due to perceived risks and uncertainty (Ullah et al., 2020). Digital literacy mitigates these concerns by improving access to information on optimal input use and the environmental and economic benefits of reduction, thereby primarily preventing overuse rather than prompting abrupt behavioral shifts. Importantly, while modest in magnitude, these marginal effects have practical relevance for China's smallholder-dominated agricultural system. At the national level, the 4.3% reduction in incremental fertilizer use probability contributes to gradual progress toward China's fertilizer reduction strategy goals. Given the strong path dependence of farmers' fertilization decisions, these statistically significant effects represent a positive and steady behavioral adjustment, with cumulative long-term impacts that could be amplified as digital literacy continues to improve.

6 Conclusions and policy implications

6.1 Conclusion

In the context of the digital economy, the rapid development of digital technology and the gradual improvement of digital literacy are reshaping farmers' decision-making processes, behavioral patterns, and agricultural production practices. Grounded in the MOA framework, this study develops a theoretical model to unpack how digital literacy may exert an influence on farmers' chemical fertilizer use decisions. Using data from 1,581 farmers from the CRRS, we employed a Gologit2 model and a two-step estimation method to empirically test the proposed relationships, while rigorously performing robustness checks and addressing endogeneity concerns.

The main findings reveal several key insights. First, this study reveals a significant negative correlation between digital literacy and farmers' chemical fertilizer use. This relationship remains robust across multiple sensitivity tests, and our instrumental variable estimation further alleviates endogeneity concerns, lending support to a cautious causal interpretation of this negative association. Second, mechanism analyses reveal that digital literacy exerts its influence on fertilizer use through four potential pathways aligned with the MOA framework: enhancing e-commerce participation (economic motivation), raising food safety awareness (social motivation), promoting the adoption of agricultural mechanization services (opportunity), and facilitating cooperatives participation (ability). Third, marginal effect analysis indicates that a one-unit increase in digital literacy is associated with a 2.3% higher probability of reducing chemical fertilizer usage and a 4.3% lower probability of increasing usage, underscoring its asymmetric yet substantive impact on farmers' input decisions.

6.2 Policy implications

Based on the empirical findings, the following policy recommendations are proposed to leverage digital literacy in promoting sustainable agricultural practices.

Enhance farmers' digital literacy. Government agencies should prioritize improving digital infrastructure in underserved rural areas to ensure equitable access to the internet and digital devices. Training programs should be developed to equip farmers with practical digital skills—including the use of agricultural apps, online marketplaces, and precision farming tools—through hands-on workshops, online courses, and extended agricultural extension services. Financial incentives, such as subsidies or tax breaks, could further encourage the adoption of digital technologies.

Promote e-commerce engagement. Specialized training programs should be introduced to help farmers leverage e-commerce platforms for selling agricultural products. This includes educating farmers on online marketing strategies, such as live streaming and digital branding, to better access markets that value green and sustainably produced goods.

Strengthen awareness of sustainable practices. Governments should utilize digital media to disseminate information on the environmental and health impacts of excessive fertilizer use. Campaigns aimed at enhancing food safety and ecological awareness can help foster intrinsic motivation among farmers to adopt greener production methods.

Integrate agricultural mechanization services with digital platforms. Developing online platforms that connect farmers with mechanization service providers can improve access to precision farming tools. Such services can reduce subjective overuse of fertilizers by offering standardized, high-precision application technologies.

Leverage agricultural cooperatives. Policies should encourage smallholder farmers to join cooperatives, which can provide technical training, resources, and peer learning opportunities focused on sustainable practices. Cooperatives can also amplify the benefits of digital literacy by facilitating knowledge sharing and collective action.

6.3 Limitations

This study has several limitations. First, the cross-sectional nature of the data limits our ability to establish causal relationships dynamically. Future research could employ panel data to capture temporal changes. Second, the measurement of digital literacy, though comprehensive, may not fully capture all relevant dimensions. Future studies could develop more nuanced indicators. Third, the generalizability of the findings may be constrained by regional characteristics; thus, applying the MOA framework in diverse contexts would be valuable.

Data availability statement

Publicly available datasets were analyzed in this study. This data can be found at: http://rdi.cssn.cn/dcsj/.

Author contributions

ZL: Writing – original draft, Writing – review & editing, Methodology, Conceptualization, Supervision, Funding acquisition, Formal analysis, Visualization, Project administration. JX: Conceptualization, Validation, Methodology, Formal analysis, Writing – review & editing. RF: Writing – original draft, Methodology.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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References

Andrews, J. C. (1988). Motivation, ability, and opportunity to process information: conceptual and experimental manipulation issues. *Adv. Consum. Res.* 15:129.

Batra, R., and Ray, M. L. (1986). Situational effects of advertising repetition: the moderating influence of motivation, ability, and opportunity to respond. *J. Consum. Res.* 12, 432–445. doi: 10.1086/208528

Baumhof, R., Decker, T., Röder, H., and Menrad, K. (2018). Which factors determine the extent of house owners' energy-related refurbishment projects? A motivation-opportunity-ability approach. *Sustain. Cities Soc.* 36, 33–41. doi: 10.1016/j.scs.2017.09.025

Chandio, A. A., Ozdemir, D., Gokmenoglu, K. K., Usman, M., and Jiang, Y. (2024). Digital agriculture for sustainable development in China: the promise of computerization. *Technol. Soc.* 76:102479. doi: 10.1016/j.techsoc.2024.102479

Chen, X., Xing, L., Wang, K., Zhang, Y., and Han, X. (2023). Nonlinear effects of internet development on chemical fertilizer usage intensity: macro evidence from China. *J. Clean. Prod.* 386:135794. doi: 10.1016/j.jclepro.2022.135794

Chen, D., Xu, T., Qiao, D., and Liu, Z. (2025). Exploring the impact of digital literacy and policy cognition on rural residents' eco-friendly behaviors. *Environ. Manag.* 75, 806–821. doi: 10.1007/s00267-024-02065-2

Chen, B., and Zhao, C. (2021). Poverty reduction in rural China: does the digital finance matter? *PLoS One* 16:e0261214. doi: 10.1371/journal.pone.0261214

Cheng, C., Gao, Q., Ju, K., and Ma, Y. (2024). How digital skills affect farmers' agricultural entrepreneurship? An explanation from factor availability. *J. Innov. Knowl.* 9:100477. doi: 10.1016/j.jik.2024.100477

Choruma, D. J., Dirwai, T. L., Mutenje, M., Mustafa, M., Chimonyo, V. G. P., Jacobs-Mata, I., et al. (2024). Digitalisation in agriculture: a scoping review of technologies in practice, challenges, and opportunities for smallholder farmers in sub-saharan africa. *J. Agric. Food Res.* 18:101286. doi: 10.1016/j.jafr.2024.101286

Cui, X., Lai, V. S., Lowry, P. B., and Lei, Y. (2020). The effects of bidder factors on online bidding strategies: a motivation-opportunity-ability (MOA) model. *Decis. Support. Syst.* 138:113397. doi: 10.1016/j.dss.2020.113397

Feng, C., Jiang, Z., Zhang, Y., Liu, X., Zhu, Y., and Liu, B. (2025). The impact and mechanism of action of the digital economy on agricultural new quality productive forces: a discussion of the threshold effect and spatial spillover effect. *Front. Sustain. Food Syst.* 9:1599265. doi: 10.3389/fsufs.2025.1599265

Ferrari, A. (2013). *DIGCOMP: A Framework for Developing and Understanding digital Competence in Europe*. Luxembourg: European Commission.

Gilster, P. (1997). *Digital Literacy*. New York: Wiley Computer Public.

Grant, J. H., Scharpenberg, D., and Manning, L. (2026). Algorithm aversion in agricultural decision-making: trust dynamics, barriers, and fertiliser-related decision support. *Agric. Syst.* 233:104630. doi: 10.1016/j.agsy.2025.104630

Guenzi, P., and Nijssen, E. J. (2023). The relationship between digital solution selling and value-based selling: a motivation-opportunity-ability (MOA) perspective. *Eur. J. Mark.* 57, 745–770. doi: 10.1108/EJM-11-2021-0907

Hu, X., and Meng, H. (2023). Digital literacy and green consumption behavior: exploring dual psychological mechanisms. *J. Consum. Behav.* 22, 272–287. doi: 10.1002/cb.2122

Ji, X., Xu, J., and Zhang, H. (2023). Environmental effects of rural e-commerce: a case study of chemical fertilizer reduction in China. *J. Environ. Manag.* 326:116713. doi: 10.1016/j.jenvman.2022.116713

Jiang, T. (2022). Mediating effects and moderating effects in causal inference empirical research. *China Ind. Econ.* 5, 100–120. doi: 10.19581/j.cnki.ciejournal.2022.05.005

Johnston, C., McDonald, J., and Quist, K. (2020). A generalized ordered probit model. *Commun. Stat. Theory Methods* 49, 1712–1729. doi: 10.1080/03610926.2019.1565780

Kumar, A., Thakur, S., and Kumar, R. (2026). Digital convergence in Agri-tech: reviewing the role of advanced technologies in enabling precision and sustainability. *Agron. J.* 118:e70310. doi: 10.1002/agj2.70310

Lei, L., Dakuan, Q., Jin, T., Lishuang, W., Yuying, L., and Xinhong, F. (2024). Research on the influence of education and training of farmers' professional cooperatives on the willingness of members to green production—perspectives based on time, method and content elements. *Environ. Dev. Sustain.* 26, 987–1006. doi: 10.1007/s10668-022-02744-2

Li, B., and Gao, Y. (2024). Impact and transmission mechanism of digital economy on agricultural energy carbon emission reduction. *Int. Rev. Econ. Finance* 95:103457. doi: 10.1016/j.iref.2024.103457

Li, R., and Gao, D. (2025). The impact of the reform of rural collective property rights system on villagers' public participation: an empirical study based on CRRS 2020 data. *PLoS One* 20:e0316899. doi: 10.1371/journal.pone.0316899

Li, B., Qiao, Y., and Yao, R. (2023). What promote farmers to adopt green agricultural fertilizers? Evidence from 8 provinces in China. *J. Clean. Prod.* 426:139123. doi: 10.1016/j.jclepro.2023.139123

Li, X., Wu, L., Gao, H., and Hu, N. (2024). Can digital literacy improve organic fertilizer utilization rates?: empirical evidence from China. *Environ. Dev. Sustain.* 26, 31921–31946. doi: 10.1007/s10668-024-04793-1

Li, D., Xu, X., Chen, C. F., and Menassa, C. (2019). Understanding energy-saving behaviors in the American workplace: a unified theory of motivation, opportunity, and ability. *Energy Res. Soc. Sci.* 51, 198–209. doi: 10.1016/j.erss.2019.01.020

Li, B., Xu, C., Zhu, Z., and Kong, F. (2021). Does e-commerce drive rural households engaged in nontimber forest product operations to adopt green production behaviors? *J. Clean. Prod.* 320:128855. doi: 10.1016/j.jclepro.2021.128855

Liu, H., Chen, Z., Wen, S., Zhang, J., and Xia, X. (2025). Impact of digital literacy on farmers' adoption behaviors of green production technologies. *Agriculture* 15:303. doi: 10.3390/agriculture15030303

Liu, X., Wang, Z., and Han, X. (2024). The impact of digital literacy on farmers' green production behavior: mediating effects based on ecological cognition. *Sustainability* 16:7507. doi: 10.3390/su16177507

Lu, S., Sun, Z., and Huang, M. (2024). The impact of digital literacy on farmers' pro-environmental behavior: an analysis with the theory of planned behavior. *Front. Sustain. Food Syst.* 8:1432184. doi: 10.3389/fsufs.2024.1432184

Ma, W., and Zheng, H. (2022). Heterogeneous impacts of information technology adoption on pesticide and fertiliser expenditures: evidence from wheat farmers in China. *Aust. J. Agric. Resour. Econ.* 66, 72–92. doi: 10.1111/1467-8489.12446

Ma, Q., Zheng, S., and Deng, P. (2022). Impact of internet use on farmers' organic fertilizer usage behavior under the climate change context: the role of social network. *Land* 11:1601. doi: 10.3390/land11091601

MacInnis, D. J., and Jaworski, B. J. (1989). Information processing from advertisements: toward an integrative framework. *J. Mark.* 53, 1–23. doi: 10.1177/002224298905300401

Mattar, J., Ramos, D. K., and Lucas, M. R. (2022). DigComp-based digital competence assessment tools: literature review and instrument analysis. *Educ. Inf. Technol.* 27, 10843–10867. doi: 10.1007/s10639-022-11034-3

Mei, L., Zheng, Y., Tian, M., and Wu, Y. (2024). Driven by the policy or bent by the market? Cracking the digital transformation code of farmer cooperatives. *Technol. Forecast. Soc. Change* 208:123659. doi: 10.1016/j.techfore.2024.123659

Miine, L. K., Akorsu, A. D., Boampong, O., and Bukari, S. (2023). Drivers and intensity of adoption of digital agricultural services by smallholder farmers in Ghana. *Heliyon* 9:e22876. doi: 10.1016/j.heliyon.2023.e23023

Morepje, M. T., Sithole, M. Z., Msweli, N. S., and Agholor, A. I. (2024). The influence of E-commerce platforms on sustainable agriculture practices among smallholder farmers in sub-Saharan Africa. *Sustainability* 16:6496. doi: 10.3390/su16156496

Ölander, F., and Thøgersen, J. (1995). Understanding of consumer behaviour as a prerequisite for environmental protection. *J. Consum. Policy* 18, 345–385. doi: 10.1007/BF01024160

Puigarnau, J. A. (2018). *Proposal for a Council Recommendation on Key Competences for Lifelong Learning*. Luxembourg: European Commission.

Qiu, H., Huang, Y., Zhang, W., He, B., Yuan, R., and Wang, Z. (2025). E-commerce operation empowers green agriculture: implication for the reduction of farmers' fertilizer usage. *Front. Sustain. Food Syst.* 9:1557224. doi: 10.3389/fsufs.2025.1557224

Qiu, H., Tang, W., Huang, Y., Deng, H., Liao, W., and Ye, F. (2024). E-commerce operations and technology perceptions in promoting farmers' adoption of green production technologies: evidence from rural China. *J. Environ. Manag.* 370:122628. doi: 10.1016/j.jenvman.2024.122628

Qu, Z., Zhang, J., Zeng, M., Zhang, K., Xu, D., Qi, Y., et al. (2024). Impact of heavy metal hazard perceptions on pollution control intentions: empirical evidence from rice farmers in China. *J. Clean. Prod.* 475:143655. doi: 10.1016/j.jclepro.2024.143655

Raimondo, M., Spina, D., Hamam, M., D'Amico, M., and Caracciolo, F. (2024). Intrinsic motivation strongly affects the readiness toward circular food consumption: evidence from the motivation–opportunity–ability model. *Br. Food J.* 126, 715–737. doi: 10.1108/BFJ-09-2022-0800

Sarku, R., and Kranjac-Berisavljevic, G. (2025). Just digital green economy transition in Ghana's smallholder agricultural sector: a systematic review. *Cogent Food Agric.* 11:2588880. doi: 10.1080/23311932.2025.2588880

Siemsen, E., Roth, A. V., and Balasubramanian, S. (2008). How motivation, opportunity, and ability drive knowledge sharing: the constraining-factor model. *J. Oper. Manag.* 26, 426–445. doi: 10.1016/j.jom.2007.09.001

Soma, T., Li, B., and Maclaren, V. (2021). An evaluation of a consumer food waste awareness campaign using the motivation opportunity ability framework. *Resour. Conserv. Recycl.* 168:105313. doi: 10.1016/j.resconrec.2020.105313

Staiger, D., and Stock, J. H. (1997). Instrumental variables regression with weak instruments. *Econometrica* 65, 557–586. doi: 10.2307/2171753

Su, M., Heerink, N., Oosterveer, P., and Feng, S. (2022). Upscaling farming operations, agricultural mechanization and chemical pesticide usage: a macro-analysis of Jiangsu Province, China. *J. Clean. Prod.* 380:135120. doi: 10.1016/j.jclepro.2022.135120

Ullah, A., Arshad, M., Kächele, H., Zeb, A., Mahmood, N., and Müller, K. (2020). Socio-economic analysis of farmers facing asymmetric information in inputs markets: evidence from the rainfed zone of Pakistan. *Technol. Soc.* 63:101405. doi: 10.1016/j.techsoc.2020.101405

van Geffen, L., van Herpen, E., Sijtsema, S., and van Trijp, H. (2020). Food waste as the consequence of competing motivations, lack of opportunities, and insufficient abilities. *Resour. Conserv. Recycl. X* 5:100026. doi: 10.1016/j.rcrx.2019.100026

Wang, X., Chen, J., and Du, X. (2024). Understanding the impact of internet access on farmers' willingness to participate in farmer professional cooperatives. *Agric. Econ.* 70, 349–361. doi: 10.17221/69/2024-agricecon

Wang, J., and Long, F. (2025). A study on the impact of farmers' digital literacy on the green transition of food production and its mechanisms. *Pol. J. Environ. Stud.* 35, 1799–1816. doi: 10.15244/pjoes/200211

Wang, X., and Zhao, X. (2025). The moderating effect of training, subsidies and propaganda on the relationship between psychological factors and farmers' willingness to reduce chemical fertilizer usage: evidence from dryland farming areas of China. *Agric. Syst.* 224:104257. doi: 10.1016/j.agsy.2024.104257

Williams, R. (2006). Generalized ordered logit/partial proportional odds models for ordinal dependent variables. *The Stata Journal: Promoting communications on statistics and Stata* 6, 58–82. doi: 10.1177/1536867x0600600104

Yang, C., Ji, X., Cheng, C., Liao, S., Obuobi, B., and Zhang, Y. (2024). Digital economy empowers sustainable agriculture: implications for farmers' adoption of ecological agricultural technologies. *Ecol. Indic.* 159:111723. doi: 10.1016/j.ecolind.2024.111723

Yu, L., Wang, W., Cui, Y., Zhou, W., Fu, Z., and He, L. (2023). Influence of capital endowment on rural households' willingness to pay for rural human settlement improvement: evidence from rural China. *Appl. Econ.* 55, 3980–3995. doi: 10.1080/00036846.2022.2120960

Zeng, Y., Tian, Y., He, K., and Zhang, J. (2020). Environmental conscience, external incentives and social norms in rice farmers' adoption of pro-environmental agricultural practices in rural Hubei province, China. *Environ. Technol.* 41, 2518–2532. doi: 10.1080/09593330.2019.1574907

Zhang, J. (2025). Unveiling the mechanism of digital literacy on job quality: Micro evidence from China. *Cities* 166:106281. doi: 10.1016/j.cities.2025.106281

Zhang, S., Li, F., and Xiao, J. (2020). Internet penetration and consumption inequality in China. *Int. J. Consum. Stud.* 44, 407–422. doi: 10.1111/ijcs.12575

Zheng, W., Luo, B., and Hu, X. (2020). The determinants of farmers' fertilizers and pesticides use behavior in China: an explanation based on label effect. *J. Clean. Prod.* 272:123054. doi: 10.1016/j.jclepro.2020.123054

Zheng, Y. Y., Zhu, T. H., and Wei, J. I. A. (2022). Does internet use promote the adoption of agricultural technology? Evidence from 1449 farm households in 14 Chinese provinces. *J. Integr. Agric.* 21, 282–292. doi: 10.1016/S2095-3119(21)63750-4

Zhou, H., and Duan, Y. (2022). Online channel structures for green products with reference greenness effect and consumer environmental awareness (CEA). *Comput. Ind. Eng.* 170:108350. doi: 10.1016/j.cie.2022.108350

Zhu, Q., van der Voort, M., Ren, G., and Bai, J. (2023). Impact of information and communication technologies on fertilizer and pesticide use efficiency of China's grain production. *Rev. Dev. Econ.* 27, 2331–2356. doi: 10.1111/rode.13039